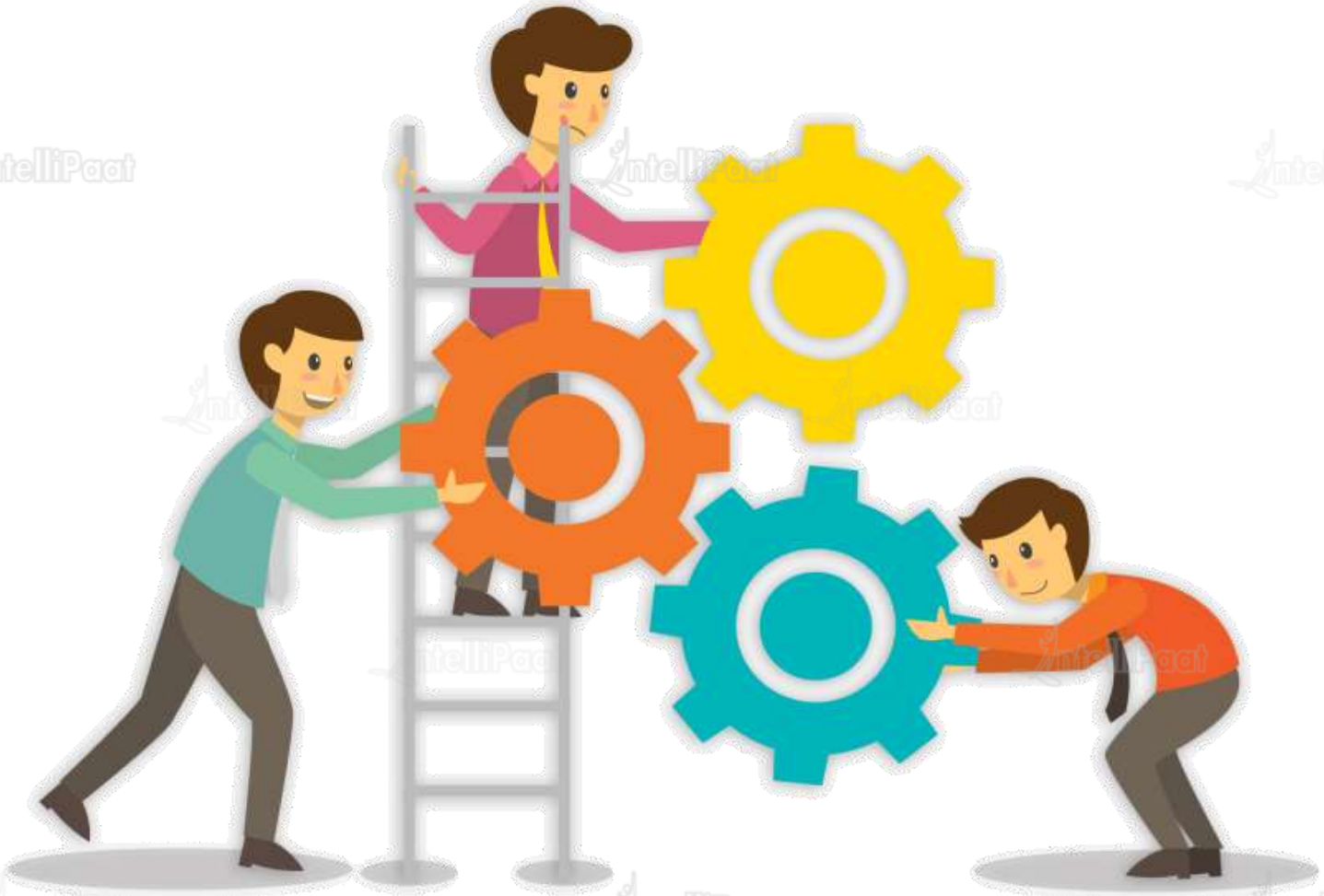
Python Training

Python Modules



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Creating a Module

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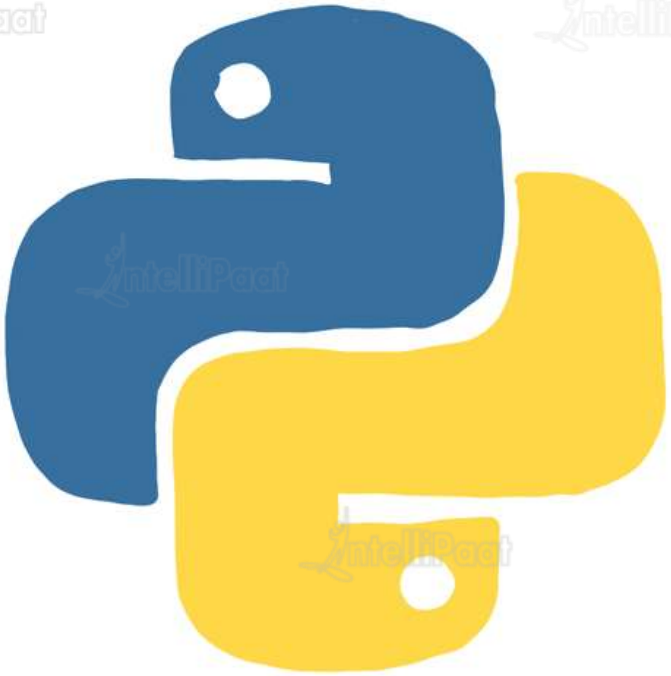
Python Module

- Consider a module to be the same as a code library.
- A file containing a set of functions you want to include in your application.
- A module allows you to logically organize your Python code.
- Grouping related code into a module makes the code easier to understand and use.
- A module is a Python object with arbitrarily named attributes that you can bind.
- Simply, a module is a file consisting of Python code.



Modules

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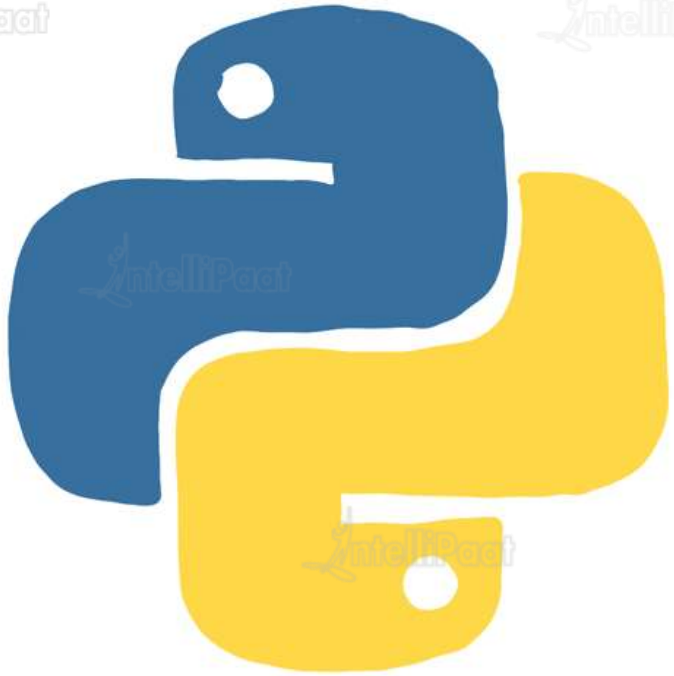
Python Module

- A module is a file containing Python definitions and statements.
- A module can define functions, classes and variables. A module can also include runnable code.
- Grouping related code into a module makes the code easier to understand and use.
- Example:-

```
# A simple module, calc.py
def add(x, y):
return (x+y)
def subtract(x, y):
return (x-y)
```



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How to import Modules in Python?

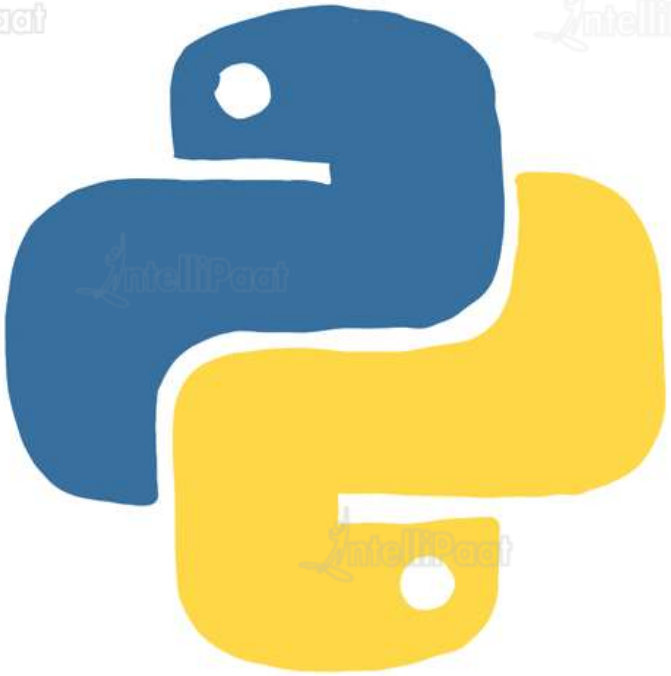
- We can import the definitions inside a module to another module or the interactive interpreter in Python.

- We use the import keyword to do this. To import our previously defined module example we type the following in the Python prompt.
  >>> import example

- This does not enter the names of the functions defined in example directly in the current symbol table.

- It only enters the module name example there.

- Using the module name we can access the function using the dot . operator. For example:
  >>> example.add(4,5.5)
  9.5
- Standard modules can be imported the same way as we import our user-defined modules.



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The Import statement

- You can use any Python source file as a module by executing an import statement in some other Python source file.

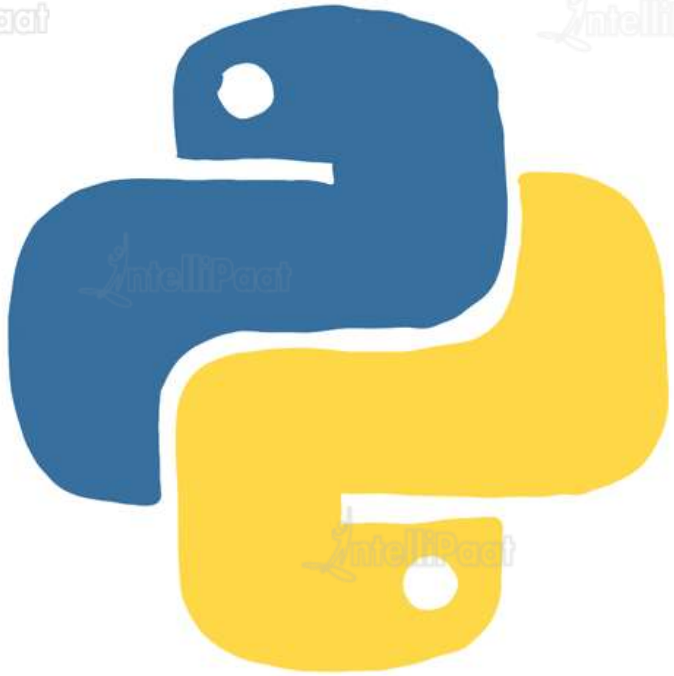
- The import has the following syntax –
- import module1[, module2[,... moduleN]

- When the interpreter encounters an import statement, it imports the module if the module is present in the search path.

- A search path is a list of directories that the interpreter searches before importing a module. For example, to import the module support.

- You need to put the following command at the top of the script –

```
#!/usr/bin/python
# Import module support
import support
# Now you can call defined function that module as follows
support.print_func("Zara")
```



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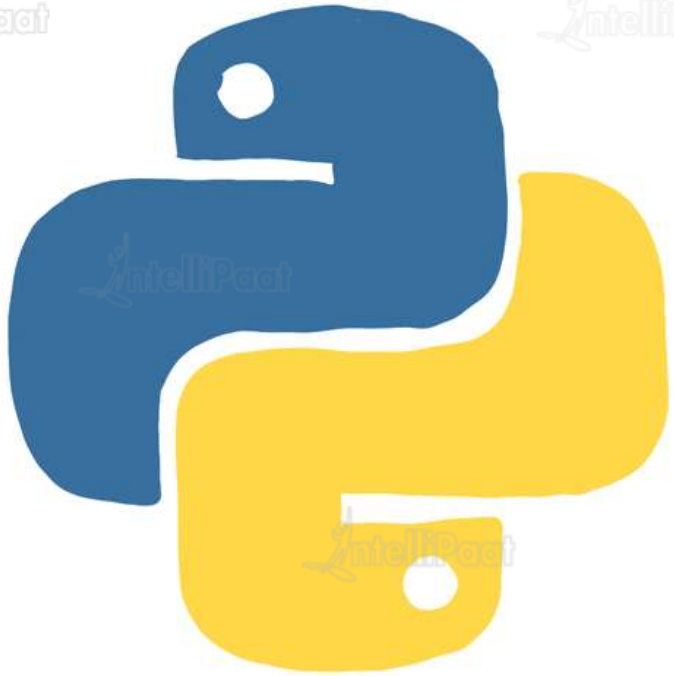
The from… Import statement

- Python from statement lets you import specific attributes from a module into the current namespace.
- The from...import has the following syntax −
  from modname import name1[, name2[, ... nameN]]
- For example, to import the function fibonacci from the module fib, use the following statement −
  from fib import fibonacci
- This statement does not import the entire module fib into the current namespace; it just introduces the item fibonacci from the module fib into the global symbol table of the importing module.
- Python from statement lets you import specific attributes from a module.
- This provides an easy way to import all the items from a module into the current namespace; however, this statement should be used sparingly.



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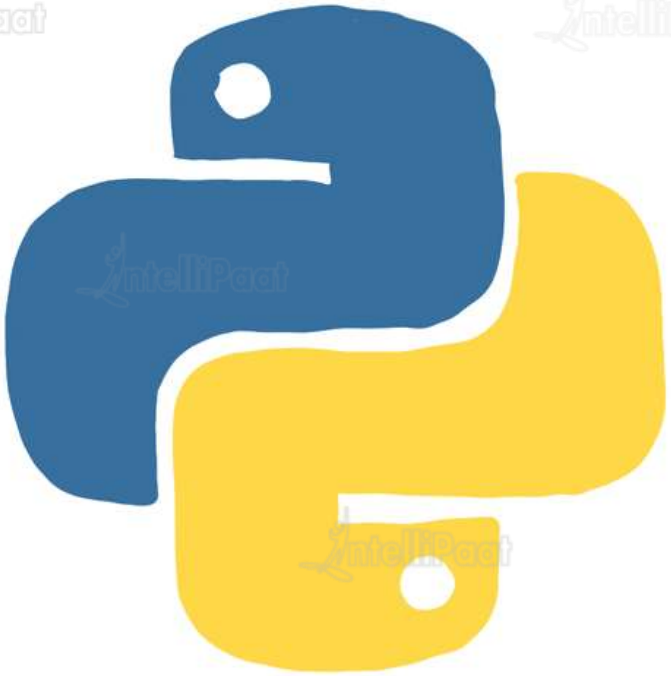
Locating Module

- When you import a module, the Python interpreter searches for the module in the following sequences −
- The current directory.
- If the module isn't found, Python then searches each directory in the shell variable PYTHONPATH.
- If all else fails, Python checks the default path. On UNIX, this default path is normally /usr/local/lib/python/.
- The module search path is stored in the system module sys as the sys.pathvariable.
- The sys.path variable contains the current directory, PYTHONPATH, and the installation-dependent default.



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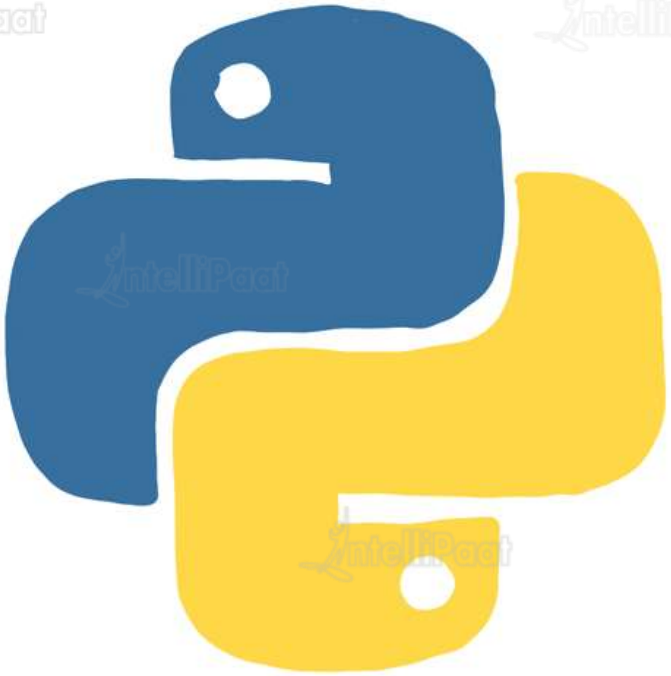
Namespace and Scoping

- Variables are names (identifiers) that map to objects.
- A namespace is a dictionary of variable names (keys) and their corresponding objects (values).
- A Python statement can access variables in a local namespace and in the global namespace.
- If a local and a global variable have the same name, the local variable shadows the global variable.
- Each function has its own local namespace.
- Class methods follow the same scoping rule as ordinary functions.
- Python makes educated guesses on whether variables are local or global.
- It assumes that any variable assigned a value in a function is local.



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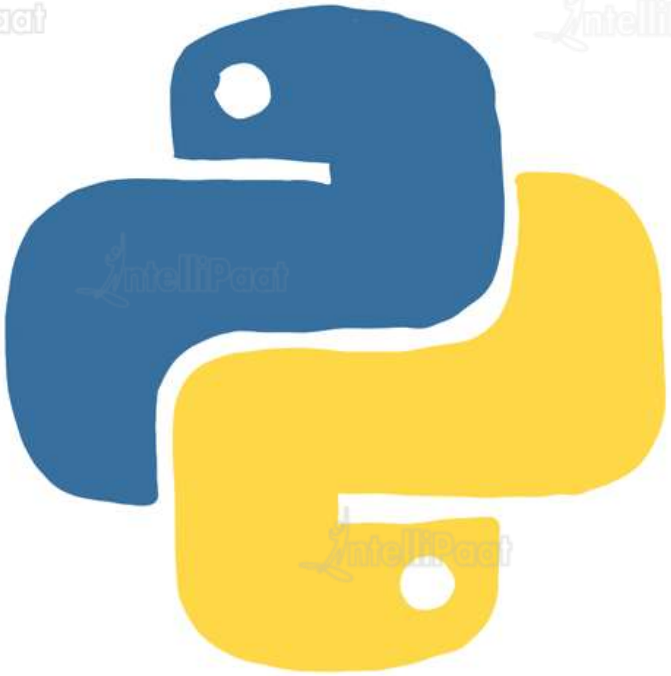
Namespace and Scoping

- Therefore, in order to assign a value to a global variable within a function, you must first use the global statement.
- The statement global VarName tells Python that VarName is a global variable.
- Python stops searching the local namespace for the variable.
- For example, we define a variable Money in the global namespace.
- Within the function Money, we assign Money a value, therefore Python assumes Money As a local variable.
- However, we assessed the value of the local variable Money Before setting it, so an UnboundLocalError is the result.
- Uncommenting the global statement fixes the problem.



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Python module search path

- While importing a module, Python looks at several places.
- Interpreter first looks for a built-in module then (if not found) into a list of directories defined in sys.path.
- The search is in this order.
- The current directory.
- PYTHONPATH (an environment variable with a list of directory).
- The installation-dependent default directory.
- We can add modify this list to add our own path.

```
>>> import sys
>>> sys.path
['',
'C:\\Python33\\Lib\\idlelib',
'C:\\Windows\\system32\\python33.zip',
'C:\\Python33\\DLLs',
'C:\\Python33\\lib',
'C:\\Python33',
'C:\\Python33\\lib\\site-packages']
```

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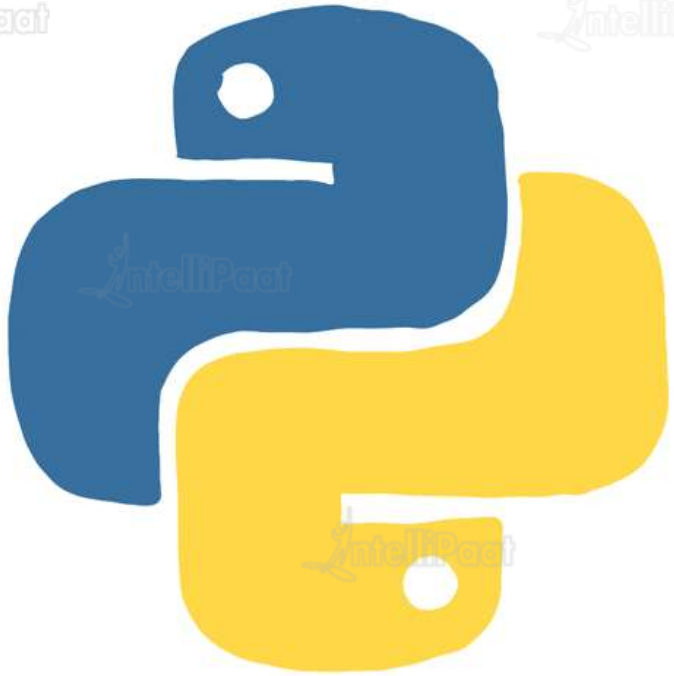
The global() and local() fucntion

- The globals() and locals() functions can be used to return the names in the global and local namespaces depending on the location from where they are called.
- If locals() is called from within a function, it will return all the names that can be accessed locally from that function.
- If globals() is called from within a function, it will return all the names that can be accessed globally from that function.
- The return type of both these functions is dictionary. Therefore, names can be extracted using the keys() function.
- The globals() function returns a dictionary containing the variables defined in theglobal namespace.
- When globals() is called from a function or method, it returns the dictionary representing the global namespace of the module.



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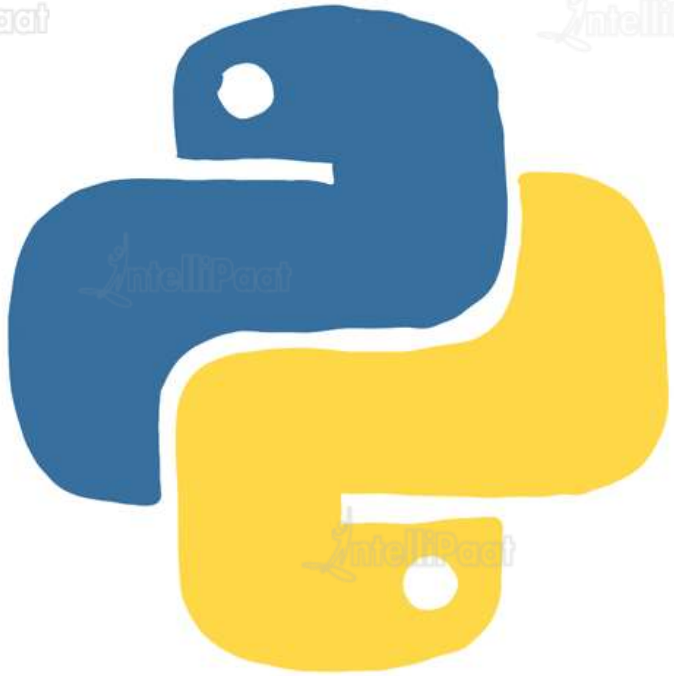
The reload functions

- The Python interpreter imports a module only once during a session. This makes things more efficient.
- When the module is imported into a script, the code in the top-level portion of a module is executed only once.
- Therefore, if you want to re execute the top-level code in a module, you can use the reload() function.
- The reload() function imports a previously imported module again. The syntax of the reload() function is this –
- reload(module_name)
- Here, module_name is the name of the module you want to reload and not the string containing the module name.
- For example, to reload hello module, do the following –
- reload(hello)



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